Dimitri Chrysafis

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Education

University of Wisconsin–Madison August 2025 - May 2027
Bachelor of Science in Computer Science
Relevant Coursework: COMP SCI 577 (Algorithms), COMP SCI 524 (Optimization), COMP SCI 400 (Programming III), COMP SCI 300 (Programming II), COMP SCI 252 (Computer Engineering), MATH 435 (Cryptography), MATH 331 (Probability), MATH 340 (Matrix and Linear Algebra)

Technical Skills

Languages: Python, C++, Swift/SwiftUI, Julia, Go, JavaScript
ML/GPU: PyTorch, WebGPU/WGSL, WebGL/GLSL, Metal, inference, quantization, model serving

Experience

Quantum Circuit Routing [Final Report] Jul 2026

- Built an exact Julia/JuMP/HiGHS MILP that routes logical circuits onto sparse superconducting hardware via SWAPs, with calibrated coupling and readout errors
- Solved all 24 experiment instances to proven optimality; a 3×3 grid topology cut routing cost **56.1%** versus the 7-qubit heavy-hex baseline

Projects

Qwen3.8-Flash-Next NVMe Inference Engine [GitHub]

- Built a disk-backed inference engine that runs the **180B-parameter Qwen3.8-Flash-Next** model on a Mac with **36 GB unified memory** by streaming MoE experts from NVMe instead of loading the full model into memory
- Implemented memory-mapped weights, dynamic expert caching, and asynchronous prefetching, and benchmarked throughput, latency, disk I/O, and peak memory

Interactive 3D Fluid Simulator (MLS-MPM) [Demo] [GitHub]

- Engineered a browser-based 3D MLS-MPM simulator in JavaScript/WebGPU, keeping particle and grid state in GPU storage buffers and simulating **400,000 particles** with staged compute kernels
- Implemented fixed-point atomic accumulation for particle-to-grid transfers plus interactive dam-break, piston, particle-injection, and orbit-camera controls

Zwift Shifter (macOS Virtual Gear Shifter) [GitHub] [Blog]

- Reverse-engineered Zwift's Click v2 Bluetooth path (FC82 service, 0x23 button frames) by tracing its Objective-C bridge with LLDB, then injected a dylib that calls Zwift's own shift logic with synthetic frames
- Shipped a SwiftUI app with status light and arrow controls that changes the real Zwift HUD gear over a local Unix socket, verified live on an Apple Silicon Mac

GPU Fractal Rendering Engine [GitHub] [Blog]

- Developed GPU fractal renderers in WebGL2/GLSL and Apple Metal for Newton, Kleinian, and Julia sets with interactive pan/zoom and shader-based iteration
- Implemented a Metal compute pipeline for **6,000×6,000** Julia frames with adaptive iteration counts and GPU-parallel image generation

GPU-Accelerated Sphere Packing in Arbitrary Meshes [GitHub] [Blog]

- Designed a stochastic tangent-sphere packing solver for arbitrary OBJ meshes, enforcing interior, surface-clearance, and pairwise non-overlap constraints
- Accelerated ray-mesh containment, nearest-surface queries, and candidate search with batched PyTorch tensors on MPS/CUDA; visualized results in Open3D